Aman K Gupta

AI / Machine Learning Engineer

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Summary

Final-year Computer Science student (AI & ML) who builds machine learning systems end to end — from training deepfake-detection and LLM-powered models to deploying them with Docker, Kubernetes, and cloud (AWS, GCP). Strong in Python and C++, with a focus on clean code, measuring results with real metrics, and shipping models that work in production, not just in a notebook. Takes ownership of problems and cares about building systems that scale.

Technical Skills

Languages: Python, C++, Java, C, JavaScript, TypeScript, SQL
ML / AI: PyTorch, TensorFlow, Deep Learning, Neural Networks, LLMs, RAG, Fine-tuning, Computer Vision, NLP, Anomaly Detection, Feature Engineering, Hyperparameter Tuning, Model Evaluation (Precision, Recall, F1)
MLOps & Systems: Docker, Kubernetes, CI/CD, Model Serving, Model Monitoring, REST APIs, System Design, Distributed Systems, Data Structures & Algorithms
Cloud & Data: AWS (S3), GCP, Firebase, Pandas, NumPy, SciPy, Matplotlib
Tools: Git, GitHub, OpenCV, n8n
Other: Bug Bounty, Vulnerability Analysis, Secure System Design

Work Experience

DOI Soft Tech June 2025 – August 2025
AI Research Intern *Remote*

- Built and deployed supervised and unsupervised ML models that supported internal workflows, cutting data pipeline latency by ∼40% through optimized feature engineering and inference scaling.
- Built an LLM-powered browser extension in Python and JavaScript to automate social media content generation, saving the team 10+ hours of manual work each week.
- Set up MLOps workflows with n8n and REST APIs; deployed containerized services on AWS and GCP using Docker and Kubernetes; analyzed 50K+ campaign records with SQL, Pandas, and NumPy.

Upwork March 2025 – April 2025
Freelance Backend Developer *Remote*

- Built backend systems and MVPs with Next.js, Firebase, and TypeScript, applying system design principles to cut deployment time by ∼30% and improve reliability.
- Developed and back-tested algorithmic trading strategies using real-time market data pipelines with sub-second latency.

Projects

DeepFake Detection System | *Python, TensorFlow, OpenCV, Computer Vision*

- Implemented and compared methods from 3–4 deepfake-detection research papers to build a working pipeline that flags manipulated images and video.
- Improved model robustness through hyperparameter tuning, regularization, and optimization; raised accuracy from ∼83% to ∼88%, evaluated with precision and recall.

Medical Expense Verification System | *Python, PyTorch, Isolation Forest, Regression*

- Built an ML pipeline processing 10K+ billing records to detect abnormal billing patterns and support fraud detection.
- Applied anomaly detection (Isolation Forest) and regression for cost validation, plus feature engineering and normalization, reducing false positives by ∼30%.

Education

University of Petroleum and Energy Studies (UPES) 2023 – 2027
B.Tech in Computer Science Engineering — Specialization in AI and ML *Dehradun, India*

- **Relevant Coursework:** Machine Learning, Deep Learning, Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Computer Vision.

Certifications

JP Morgan Chase Virtual SWE Internship (Forage) | Goldman Sachs SWE Job Simulation (Forage) | Walmart Advanced SWE Job Simulation (Forage)